

Mayukh Jain

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SUMMARY

AI/ML Engineer specializing in Cloud Computing and Automation with experience in deploying scalable, containerized microservices. Proficient in architecting RAG pipelines and AI-driven systems within Cloud environments (AWS, Azure). Adept at integrating AI models with edge-ready platforms and optimizing network-related application performance.

EXPERIENCE

FOSSEE Summer Fellow (Osdag)

April 2026 – Present

IIT Bombay (Remote)

- Contributing to Osdag, an open-source structural design software, specifically developing the Desktop UI for the Bridge Module.
- Utilizing Python and PySide6 (Qt) to build responsive, data-driven interfaces for complex bridge component design and design verification.
- Implementing parametric CAD visualization features by integrating backend design logic with the desktop GUI to enhance user workflow.

GirlScript Summer of Code

May 2026 – Aug 2026

Open Source Contributor (AI & Agents)

- Architected and implemented Retrieval-Augmented Generation (RAG) pipelines using Python, LangChain, and vector databases (e.g., Pinecone or ChromaDB) to enhance LLM context-awareness and reduce hallucination rates in autonomous agentic workflows.
- Developed modular agentic frameworks by integrating ReAct prompting and Tool-calling capabilities, enabling AI agents to autonomously execute complex multi-step tasks across diverse APIs and cloud environments.
- Engineered scalable open-source AI infrastructure by optimizing codebase modularity and implementing comprehensive unit testing, facilitating seamless collaboration with global maintainers and accelerating the resolution of high-priority repository issues.

TECHNICAL SKILLS

Programming Languages & Databases: Python, C++, SQL, Java, JavaScript, Vector Databases (FAISS/ChromaDB)

Cloud & MLOps: AWS (EC2, S3, Lambda, IAM), Docker, Hugging Face Spaces, Vercel, Azure, Render, Qdrant, Supabase

AI/ML & GenAI: LangChain, RAG (Retrieval Augmented Generation), TensorFlow, PyTorch, Scikit-learn, DeepSeek API (LLMs), OpenCV, Pytesseract (OCR), Hugging Face Transformers, Cosine Similarity,

Frameworks: PyTorch, FastAPI, Flask, Streamlit, React, PySide(Qt)

Libraries & Tools: Pandas, NumPy, Matplotlib, Seaborn, Git, MySQL, Jupyter Notebook, REST APIs

PROJECTS

Academia.AI | [Link](#) | [Code](#)

Jan 2026

Intelligent Research & Learning Assistant

Fast API, Supabase, RAG, Google Embeddings, Edge TTS

- Engineered a RAG-based study platform utilizing Google Gemini 1.5 and pgvector to enable context-aware Q&A across uploaded research papers with sub-second retrieval latency.
- Developed multi-modal generative features, including "CourseGenie" (an adaptive curriculum generator with interactive quizzes) and "Podcast Studio" (transforming topics into engaging 2-person audio dialogues).
- Architected a decoupled full-stack system using FastAPI and React, featuring real-time audio visualization, secure session-based history management, and split-screen document previewing for enhanced productivity.

Leaf Compass | [Link](#) | [Code](#)

July 2025 – Dec 2025

Full-Stack AgriAI App (Computer Vision & NLP)

React, Python, FastAPI, TensorFlow, Scikit-learn, LLMs

- Built a Full-Stack Agri-Tech Platform using React and FastAPI, integrating CNNs and Regression models for disease diagnosis, crop recommendation, fertilizer recommendation and yield prediction with ~90% accuracy and deployed a containerized backend on Hugging Face Spaces (Docker) with a Vercel frontend.
- Engineered a context-aware AI chatbot using the DeepSeek-V3.2 API, reducing user query resolution time by providing instant, interactive agricultural advice.

NYAYA.AI | [Link](#) | [Code](#)

Dec 2025

Legal Intelligence Platform

FastAPI, RAG, Llama 3, Pinecone, Docker, Vercel

- Developed a full-stack legal AI companion using Retrieval-Augmented Generation (RAG) to interpret the Bhartiya Nyaya Sanhita (BNS) with 98% accuracy.
- Architected a vector search pipeline using Pinecone and HuggingFace embeddings to query 15k+ precedents in sub-second time.
- Integrated Llama 3.3 (via Groq) to build an AI Verdict Predictor and Automated Document Drafter for affidavits and agreements.

EDUCATION

VIT Bhopal University | CGPA: 8.52

Bhopal, India

B.Tech. in Computer Science, Cloud Computing and Automation

Aug 2023 – Present

CERTIFICATIONS

AWS Solution Architect Associate

Smartbridge Certified Machine Learning in collaboration with Google

Applied Machine Learning in Python from University of Michigan

AWS Cloud Practitioner Essentials

Microsoft Certified: Azure Data Fundamentals

IBM Cloud Computing Fundamentals

DECLARATION : I hereby declare that the details provided above are true and correct to the best of my knowledge and belief.

Bhopal, Madhya Pradesh

Mayukh Jain